Monday LAB Questions (9-sept-2024)

1) Write a program that prints the sum of the first n odd numbers using a function.

Example: Input: Enter the value of n: 5

Output: Sum of first 5 odd numbers: 25

2) Write a program to find both the GCD and LCM of two numbers using multiple functions.

Example: Input: Enter two numbers: 18 24

Output: GCD: 6

LCM: 72

3) Write a program to convert a decimal number to its binary equivalent using functions.

Example: Input: Enter a decimal number: 10

Output: Binary equivalent: 1010

4) Write a program to find the prime factors of a number using functions.

Example: Input: Enter a number: 60

Output: Prime factors: 2, 2, 3, 5

5) Write a program to check if a number is a Harshad (Niven) number. A number is called a Harshad number if it is divisible by the sum of its digits. Use multiple functions to check divisibility and calculate the sum of digits.

Example: Input: Enter a number: 18

Output: 18 is a Harshad number (18 / (1+8) = 2)